HIERARCHICAL MODEL OF OCCT BUS ARRIVAL TIMES

Brendan Heaney

Professor Hu

Math 488S

5/4/2026

MY RESEARCH QUESTION

- Most of us have experience with the OCCT bus system and delays
- Official schedule makes the incorrect assumption that speed is uncorrelated with time
- Guiding research question is “At each stop, at each scheduled arrival, what is our best estimate as to the true mean arrival time?”

BACKGROUND/SIGNIFICANCE

SCOPE

- I aim to create a schedule with the same level of resolution as the official schedule. For this analysis, we will restrict our analysis to the Weekend Target stop

SIGNIFIANCE

- Binghamton University has almost 19,000 students, many of whom rely on OCCT

APPLICATIONS

- A more accurate schedule, would be helpful to students in planning their commutes and OCCT staff in determining scheduling

DATA SOURCES

OCCT API

- We may send a request to https://binghamtonupublic.etaspot.net/service.php
- This returns 33 pieces of data for each of the 23 buses in the system, many of which are at depot
- The OCCT API similarly can be pinged for stop and route information

Stop Information

- To compile our data we need both the true and OCCT arrival times
- Thankfully, OCCT hosts all of their bus arrival times on their website as well as uploading a Spring Master Schedule PDF
- https://occtransport.org/pages/routeschedule.html

COLLECTING DATA

I recorded two weeks of data (1.1 million lines) as follows.

DATA COLLECTION

- Every 20 Seconds I ping the OCCT API
- This produces 23 data points, containing routeID, equipmentID(For bus), and nextStopID

DATA CLEANSING GOALS

- Data Cleansing will be handled in Python with Pandas
- We aim to record how late or early each bus is at each stop

DATA CLEANSING

- We fetch a table converting routeID and nextStopID to routes and stops
- We find delays by subtracting nearest scheduled arrival time from actual arrival time

RAW DATA COLLECTED

- The OCCT API returns 23 columns of data
- Among them are poll_timestamp, equipmentID, routeID, nextStopID, which will form the basis of our data analysis

occt_2026-04-10.csv

W1 | nextStopID

	A	B	C	D	E	V	W	X	Y	Z	AA	AB	AC
1	poll_times	equipment	vehicleTyp	trainID	routeID	receiveTim	nextStopID	nextPatter	nextStopEx	lastStopID	lastPatterr	lastStopEx	nextStopPc
2	2026-04-1	37	Bus	0	777	1.78E+12	0	0		0	0		
3	2026-04-1	380	Bus	0	8	1.77E+12	0	0		0	0		
4	2026-04-1	381	Bus	0	777	1.78E+12	0	0		0	0		
5	2026-04-1	526	Bus	0	12	1.78E+12	0	0		0	0		
6	2026-04-1	527	Bus	22632	1	1.78E+12	33	10353	33	32	10352	32	26.63339
7	2026-04-1	528	Bus	0	4	1.78E+12	0	0		0	0		
8	2026-04-1	685	Bus	0	7	1.78E+12	0	0		1	8817		
9	2026-04-1	747	Bus	0	8	1.78E+12	0	0		15	8226		
10	2026-04-1	748	Bus	0	8	1.78E+12	0	0		10	8221		
11	2026-04-1	762	Bus	0	777	1.78E+12	0	0		2	8215		
12	2026-04-1	763	Bus	0	8	1.77E+12	0	0		0	0		
13	2026-04-1	834	Bus	0	12	1.78E+12	0	0		0	0		
14	2026-04-1	836	Bus	0	5	1.78E+12	0	0		1	8203		
15	2026-04-1	889	Bus	0	8	1.78E+12	0	0		15	9666		
16	2026-04-1	890	Bus	0	777	1.77E+12	0	0		0	0		
17	2026-04-1	891	Bus	0	16	1.78E+12	0	0		37	9596		
18	2026-04-1	915	Bus	0	4	1.78E+12	0	0		1	9500		
19	2026-04-1	916	Bus	0	777	1.78E+12	0	0		0	0		
20	2026-04-1	917	Bus	0	777	1.77E+12	0	0		0	0		
21	2026-04-1	972	Bus	22692	3	1.78E+12	48	9487	48	47	9486	47	10.4015
22	2026-04-1	984	Bus	0	15	1.78E+12	0	0		0	0		
23	2026-04-1	985	Bus	0	15	1.78E+12	0	0		0	0		
24	2026-04-1	986	Bus	0	15	1.78E+12	0	0		1	10161		
25	2026-04-1	37	Bus	0	777	1.78E+12	0	0		0	0		
26	2026-04-1	380	Bus	0	8	1.77E+12	0	0		0	0		
27	2026-04-1	381	Bus	0	777	1.78E+12	0	0		0	0		
28	2026-04-1	526	Bus	0	12	1.78E+12	0	0		0	0		
29	2026-04-1	527	Bus	22632	1	1.78E+12	33	10353	33	32	10352	32	61.42554
30	2026-04-1	528	Bus	0	4	1.78E+12	0	0		0	0		
31	2026-04-1	685	Bus	0	7	1.78E+12	0	0		1	8817		
32	2026-04-1	747	Bus	0	8	1.78E+12	0	0		15	8226		
33	2026-04-1	748	Bus	0	8	1.78E+12	0	0		10	8221		

occt_2026-04-10

Ready | Accessibility: Unavailable

PROBLEMS AND SOLUTIONS (PART 1)

Inter-Line Dependency

Problem: There are 16 bus lines we need to evaluate. Should there be dependency?

Solution: No, they are too different and we have a large amount of data for each line. It is not needed.

Within-Line Dependency

Problem: Buses run at different speeds at different times,

Solution: We use a hierarchical model, where bus parameters are drawn from a within-line distribution

Bus Arrival Times skew

Problem: Buses run significantly late more often than they run significantly early

Solution: We assume that buses follow a gamma distribution, allowing for skew.

PROBLEMS AND SOLUTIONS (PART 2)

Zero-bounding

Problem: Gamma likelihood is not defined below zero

Solution: Solution – We subtract the smallest delay off

Incorrect Routes

Problem: Buses occasionally will report themselves as running the wrong route

Solution: Manually verify extreme outliers

University Downtown Center

Problem: Bus drivers update their route at the exact moment of their arrival to UDC

Solution: We write a special case to catch buses switching route

MODEL FOR A GIVEN STOP

For bus i (Say, Saturday) in time j (Say, the 10:54 VS Target stop), we let

$$z_{ij} = Y_{ij} - s,$$

With s being the earliest arrival at a stop. As this is done to prevent early buses from breaking our likelihood function

We parameterize Gamma slightly strangely so we can use lambda for posterior predictive checks

$$E[Y_{ij} \mid \lambda_j, \phi] = \lambda_j + \mathrm{s}.$$

- Sampling density

$$z_{ij} \overset{i.i.d.}{\sim} \text{Gamma}\left(\phi, \frac{\phi}{\lambda_j}\right),$$

- Stage 1 Prior

$$\lambda_j \sim \text{Normal}(\mu, \tau^2)T(0, \infty).$$

$$\phi \sim \text{Gamma}(\alpha_\phi, \beta_\phi).$$

- Hyperpriors

$$\mu \sim \text{Normal}(\mu_0, g_0^2),$$

$$1/\tau^2 \sim \text{Gamma}(\alpha_\tau, \beta_\tau).$$

THE HIERARCHICAL MODEL FOR A STOP

$$\mu, \tau$$

$$\lambda_1 \qquad \dots \qquad \lambda_J$$

$$\{z_{i1}\} \qquad \dots \qquad \{z_{iJ}\}$$

$$\varphi$$

DATA CLEANSING: TARGET ON VESTAL SHOPPING SHUTTLE

We put the VSS bus arrival delays into a .csv file, matched with their scheduled time, and calculated delays in minutes from this time

	A	B	C	D	E
1	Schedule	Delay	Date	ArrivalTim	TripID
2	10:09	0.966667	4/11/2026	09:58.0	700
3	10:09	6.683333	4/12/2026	15:41.0	846
4	10:09	0.8	4/18/2026	09:48.0	700
5	10:09	6.5	4/19/2026	15:30.0	846
6	10:54	-0.36667	4/11/2026	53:38.0	701
7	10:54	0.683333	4/12/2026	54:41.0	848
8	10:54	2.133333	4/18/2026	56:08.0	701
9	10:54	2.166667	4/19/2026	56:10.0	848
10	11:39	2.3	4/11/2026	41:18.0	712
11	11:39	0.35	4/12/2026	39:21.0	850
12	11:39	2.466667	4/18/2026	41:28.0	712
13	11:39	4.166667	4/19/2026	43:10.0	850
14	12:24	0.633333	4/11/2026	24:38.0	702
15	12:24	5.683333	4/12/2026	29:41.0	853

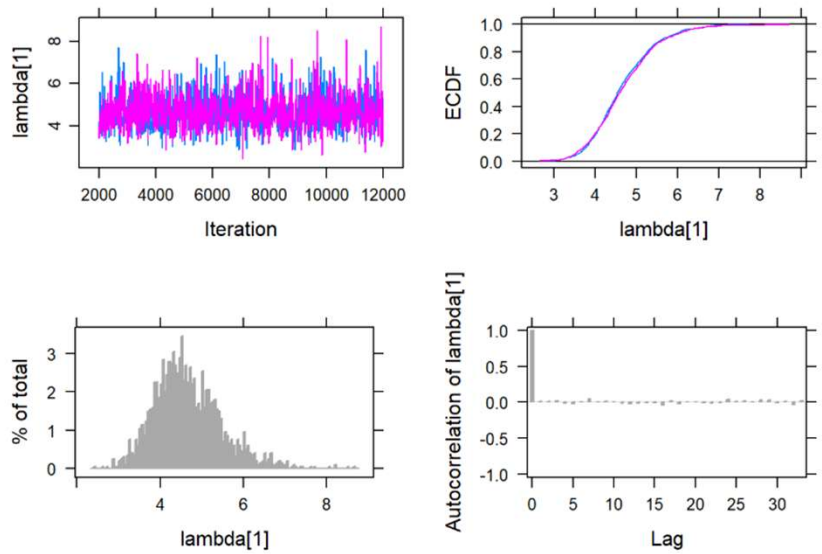
VSS WEEKEND TARGET DIAGNOSTICS: LAMBDA[1]

To make sure my model fit and everything converged I did diagnostics on the Target bus stop over the weekend

We see For lambda[1], mean delay for the first bus route in the morning, good convergence

```
plot(posterior, vars = "lambda[1]")
```

```
## Generating summary statistics and plots (these will NOT be saved for
## reuse)...
## Calculating summary statistics...
## Calculating the Gelman-Rubin statistic for 1 variables....
```

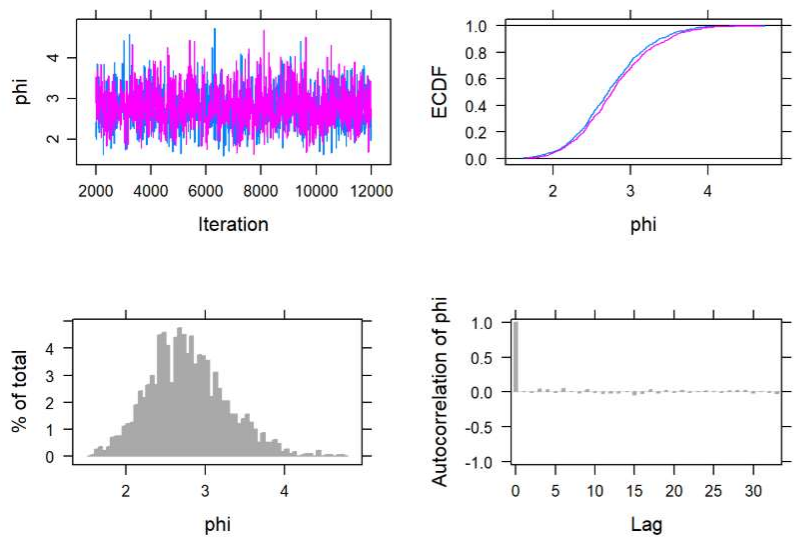


VSS WEEKEND TARGET DIAGNOSTICS: PHI

For phi, a gamma parameter, we see two chains that were able to similarly explore the sample space with little signs of distress

```
plot(posterior, vars = "phi")
```

```
## Generating summary statistics and plots (these will NOT be saved for
## reuse)...
## Calculating summary statistics...
## Calculating the Gelman-Rubin statistic for 1 variables....
```



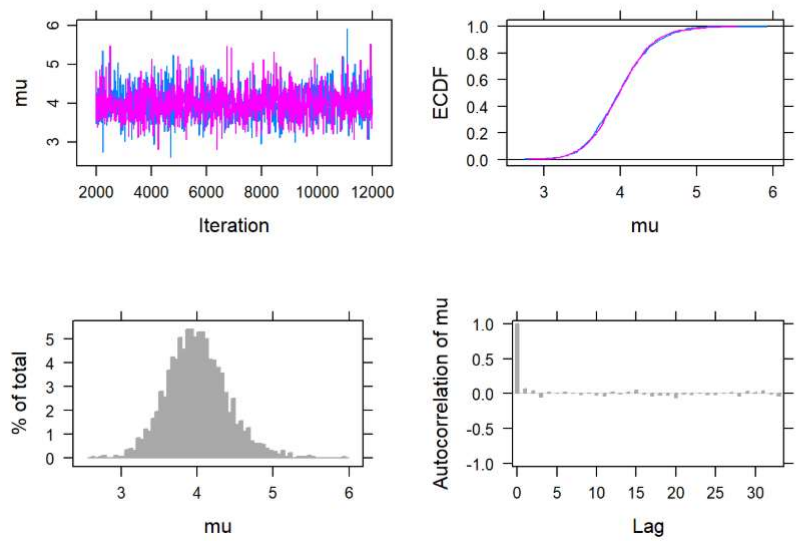
VSS WEEKEND TARGET DIAGNOSTICS: MU

We similarly see two chains with little autocorrelation able to see most of the sample space

We can interpret mu as the average delay per line

```
plot(posterior, vars = "mu")
```

```
## Generating summary statistics and plots (these will NOT be saved for
## reuse)...
## Calculating summary statistics...
## Calculating the Gelman-Rubin statistic for 1 variables....
```

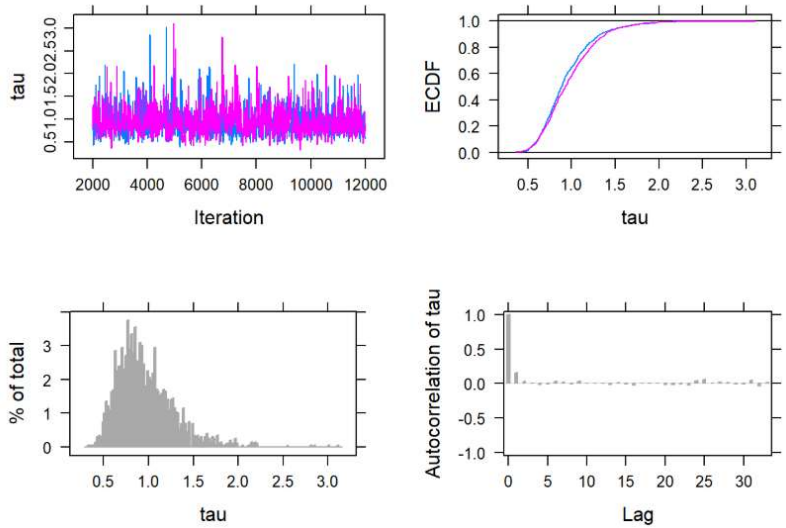


VSS WEEKEND TARGET DIAGNOSTICS: TAU

We see good exploration of the sample space, and only slight autocorrelation

```
plot(posterior, vars = "tau")
```

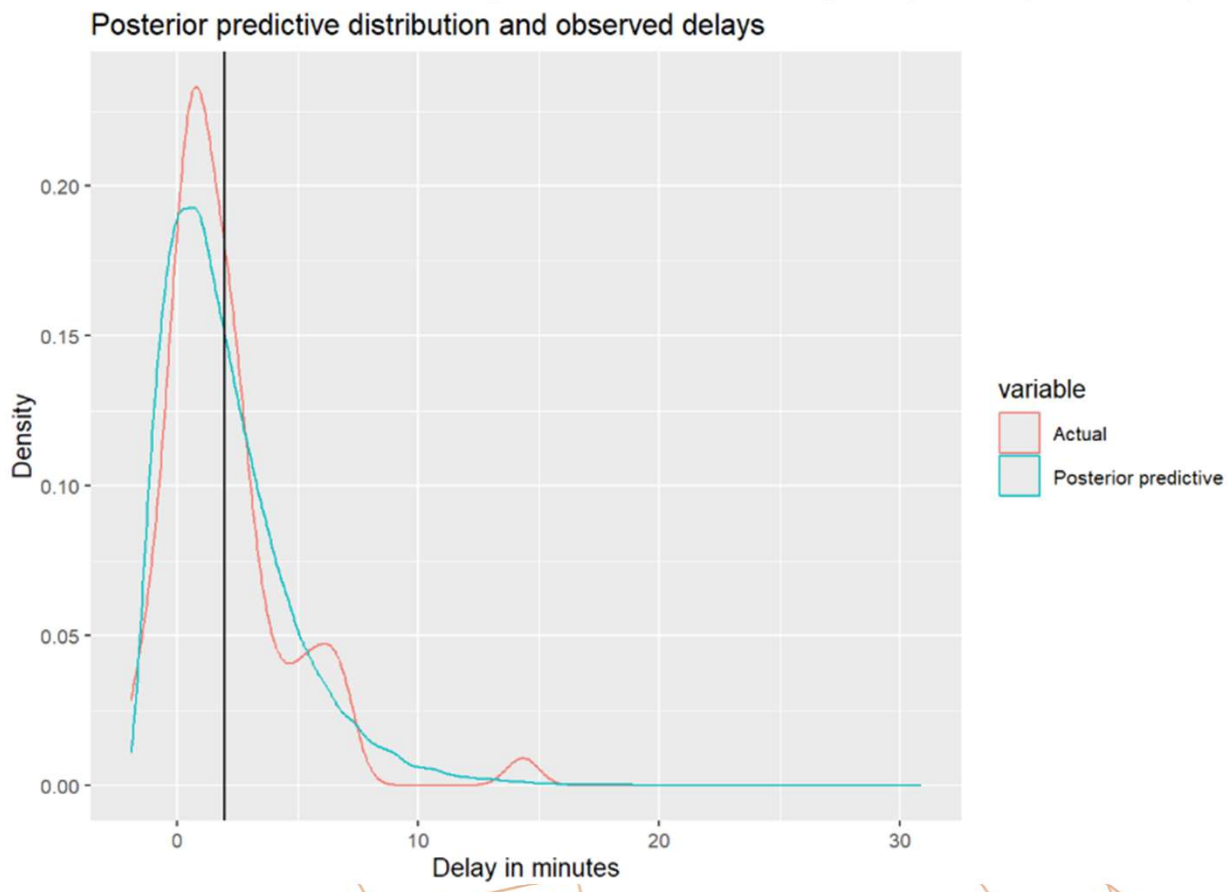
```
## Generating summary statistics and plots (these will NOT be saved for
## reuse)...
## Calculating summary statistics...
## Calculating the Gelman-Rubin statistic for 1 variables....
```



POSTERIOR PREDICTIVE CHECKS: VESTAL SHOPPING AT TARGET

Random draws from the posterior seem to align quite well with actual bus delays!

We do not notice either systemic bias or overfitting



LIMITATIONS

Limited Data

I collected data for 2 weeks, and in that time recorded 4 instances of each Weekend VSS bus arriving for 65 buses in total. Running the data scraper for longer would create better estimates

Between-Stop Correlation

In real life, information about a bus's expected arrival at one stop is contained at other stops. I do not attempt to handle this.

ESTIMATE AT ALL FOUR STOPS

Vestal Shopping WE

University Union		Target		Walmart		The Shoppes at Vestal	
10:00:00 AM	**10:02:25 AM**	10:09:00 AM	**10:12:29 AM**	10:15:00 AM	**10:16:45 AM**	10:20:00 AM	**10:20:49 AM**
10:45:00 AM	**10:42:13 AM**	10:54:00 AM	**10:55:33 AM**	11:00:00 AM	**10:59:25 AM**	11:05:00 AM	**11:04:07 AM**
11:30:00 AM	**11:26:17 AM**	11:39:00 AM	**11:41:27 AM**	11:45:00 AM	**11:46:37 AM**	11:50:00 AM	**11:50:43 AM**
12:15:00 PM	**12:16:39 PM**	12:24:00 PM	**12:26:39 PM**	12:30:00 PM	**12:32:26 PM**	12:35:00 PM	**12:37:31 PM**
1:00:00 PM	**12:58:44 PM**	1:09:00 PM	**1:11:06 PM**	1:15:00 PM	**1:15:46 PM**	1:20:00 PM	**1:20:19 PM**
1:45:00 PM	**1:42:11 PM**	1:54:00 PM	**1:55:33 PM**	2:00:00 PM	**2:00:22 PM**	2:05:00 PM	**2:06:04 PM**
2:30:00 PM	**2:31:32 PM**	2:39:00 PM	**2:42:12 PM**	2:45:00 PM	**2:47:09 PM**	2:50:00 PM	**2:52:09 PM**
3:15:00 PM	**3:16:25 PM**	3:24:00 PM	**3:25:49 PM**	3:30:00 PM	**3:30:46 PM**	3:35:00 PM	**3:35:35 PM**
4:00:00 PM	**3:58:26 PM**	4:09:00 PM	**4:11:13 PM**	4:15:00 PM	**4:16:22 PM**	4:20:00 PM	**4:20:55 PM**
4:45:00 PM	**4:49:31 PM**	4:54:00 PM	**4:56:44 PM**	5:00:00 PM	**5:02:05 PM**	5:05:00 PM	**5:06:58 PM**
5:30:00 PM	**5:32:04 PM**	5:39:00 PM	**5:40:13 PM**	5:45:00 PM	**5:45:10 PM**	5:50:00 PM	**5:49:36 PM**
6:15:00 PM	**6:16:09 PM**	6:24:00 PM	**6:25:43 PM**	6:30:00 PM	**6:31:32 PM**	6:35:00 PM	**6:36:23 PM**
7:00:00 PM	**7:01:11 PM**	7:09:00 PM	**7:09:16 PM**	7:15:00 PM	**7:14:59 PM**	7:20:00 PM	**7:18:40 PM**
7:45:00 PM	**7:45:45 PM**	7:54:00 PM	**7:56:36 PM**	8:00:00 PM	**8:00:45 PM**	8:05:00 PM	**8:04:42 PM**
8:30:00 PM	**8:27:11 PM**	8:39:00 PM	**8:39:21 PM**	8:45:00 PM	**8:43:40 PM**	8:50:00 PM	**8:46:39 PM**
9:15:00 PM	**9:14:02 PM**	9:24:00 PM	**9:28:18 PM**	9:30:00 PM	**9:32:36 PM**	9:35:00 PM	**9:35:56 PM**
Official	**Estimate**						

LINK TO SCHEDULE ESTIMATE

THANK YOU!

Brendan Heaney